

=== CONFIGURACION GNB (1779615716) ===

This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.
This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 20 MHz bandwidth
To run the srsRAN Project gNB with this config, use the following command:
sudo ./gnb -c gnb.yaml

```
cu_cp:
  amf:
    addr: 10.53.1.2          # The address or hostname of the AMF.
    port: 38412
    bind_addr: 0.0.0.0       # A local IP that the gNB binds to for traffic from the AMF. IMPORTANT: As per strict
    supported_tracking_areas:
      - tac: 7
      plmn_list:
        - plmn: "315010"     # PLMN broadcasted by the gNB (MCC 315, MNC 010).
          tai_slice_support_list:
            - sst: 4          # V2X Slice (3GPP TS 23.501, Table 5.15.2.2-1, Pag 199)
    inactivity_timer: 7200    # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [1 - 7200]

ru_sdr:
  device_driver: zmq         # The RF driver name.
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=30.72e6 # ZMQ ports for gNB TX (2000) and RX (2001)
  srate: 30.72              # RF sample rate in Msps.
  tx_gain: 75               # Transmit gain of the RF might need to adjusted to the given situation.
  rx_gain: 75               # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:
  dl_arfcn: 620000          # ARFCN of the downlink carrier (center frequency) for Band 78.
  band: 78                  # The NR band (Band 78 for dl_arfcn 620000).
  channel_bandwidth_MHz: 20 # Bandwidth in MHz. Number of PRBs will be automatically derived (106 PRBs for 20MHz)
  common_scs: 15            # Subcarrier spacing in kHz used for data. (Must match SSB SCS, if defined).
  plmn: "315010"           # PLMN broadcasted by the gNB (MCC 315, MNC 010).
  tac: 7                    # Tracking area code (needs to match the core configuration).
  pdcch:
    common:
      ss0_index: 0          # Set search space zero index to match srsUE capabilities
      coreset0_index: 12    # Set search CORESET Zero index to match srsUE capabilities (consistent with 15kHz SSB)
    dedicated:
      ss2_type: common      # Search Space type, has to be set to common
      dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
  prach:
    prach_config_index: 1   # Sets PRACH config to match what is expected by srsUE
  pdsch:
    mcs_table: qam64        # Sets PDSCH MCS to 64 QAM
  pusch:
    mcs_table: qam64        # Sets PUSCH MCS to 64 QAM

log:
  filename: /tmp/gnb.log    # Path of the log file.
  all_level: info           # Logging level applied to all layers.
  hex_max_size: 0

pcap:
  mac_enable: false         # Set to true to enable MAC-layer PCAPs.
  mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
  ngap_enable: false        # Set to true to enable NGAP PCAPs.
  ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
  e2ap_enable: true         # Set to true to enable E2AP PCAPs.
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

metrics:
  layers:
    enable_rlc: true        # Enable RLC metrics
```

```
    enable_sched: true          # Enable DU scheduler metrics
periodicity:
  du_report_period: 1000      # Set DU statistics report period to 1s
```

=== CONFIGURACION UE (1779615716) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 30.72e6 # RF sample rate in Hz, matching gNB.
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=30.72e6 # ZMQ ports for UE TX (2001) and RX (2000)

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 78 # NR band, matching gNB.
nof_carriers = 1

max_nof_prb = 106 # Max number of PRBs for 20MHz bandwidth, 15kHz SCS.
nof_prb = 106 # Number of PRBs to use.

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 315010000000001 # IMSI with MCC 315, MNC 010, matching gNB PLMN.
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER COMPOSE (1779615716) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
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# and at http://www.gnu.org/licenses/.
#

services:
  5gc:
    container_name: open5gs_5gc
    build:
      context: open5gs
      target: open5gs
      args:
        OS_VERSION: "22.04"
        OPEN5GS_VERSION: "v2.7.0"
    env_file:
      - ${OPEN_5GS_ENV_FILE:-open5gs/open5gs.env}
    privileged: true
    ports:
      - "9898:9999/tcp"
      # Uncomment port to use the 5gc from outside the docker network
      - "38412:38412/sctp"
      - "2152:2152/udp"
    command: 5gc -c open5gs-5gc.yml
    healthcheck:
      test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
      interval: 3s
      timeout: 1s
      retries: 60
    networks:
      ran:
        ipv4_address: ${OPEN5GS_IP:-10.53.1.2} # Static IP for Open5GS core.

  gnb:
    container_name: srsran_gnb
    # Build info
    image: srsran/gnb
    build:
      context: ..
      dockerfile: docker/Dockerfile
      args:
        OS_VERSION: "24.04"
      # privileged mode is required only for accessing usb devices
      privileged: true
      # Extra capabilities always required
      cap_add:
        - SYS_NICE
        - CAP_SYS_PTRACE
    volumes:
      # Access USB to use some SDRs
```

```

- /dev/bus/usb/:/dev/bus/usb/
# Sharing images between the host and the pod.
# It's also possible to download the images inside the pod
- /usr/share/uhd/images:/usr/share/uhd/images
# Save logs and more into gnb-storage
- gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current netowrk configuration creastes a private netwoek with both containers attached
# An alterantive would be `network: host`. That would expose your host network into the container. It's the easie
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.3} # Static IP for srsRAN gNB.
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
  gnb_config.yml:
    file: ./gnb.yaml # Path to your desired gNB config file (assuming it's in the same directory).
  gnb_compose_config.yml:
    content: |
      cu_cp:
        amf:
          addr: ${OPEN5GS_IP:-10.53.1.2}
          bind_addr: 0.0.0.0 # IMPORTANT: As per strict rule, bind_addr must be 0.0.0.0.
        metrics:
          autostart_stdout_metrics: true
          enable_json: true
        remote_control:
          bind_addr: 0.0.0.0
          enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:
      driver: default
      config:
        - subnet: 172.19.1.0/24

```